Ripresa da lesson 12

$\{X_i\}$ i.i.d.

x_i = observation

→ estimate empirically $\tilde{x} = \frac{1}{N} \sum_{i=1}^{N} x_i$

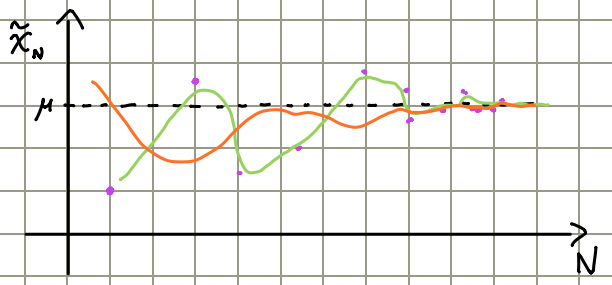
If we repeat the experiment

1st exp: $\{x_1, x_2, \dots, x_N\} \rightarrow \tilde{x}^{(1)} = \frac{1}{N} \sum_{i=1}^{N} x_i$

2nd exp: $\{x_1', x_2', \dots, x_N'\} \rightarrow \tilde{x}^{(2)} = \frac{1}{N} \sum_{i=1}^{N} x_i'$

→ $\tilde{x} = \frac{1}{N} \sum_{i=1}^{N} x_i$ Is $\tilde{x}^{(1)} \stackrel{?}{=} \tilde{x}^{(2)}$

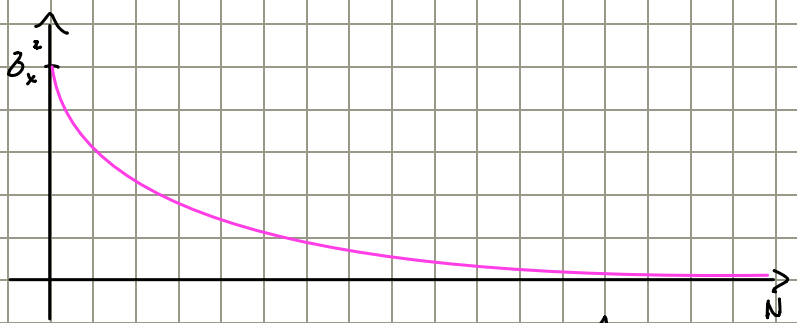
I would say no, because the obs. are r.v. so $\tilde{x}^{(1)} \neq \tilde{x}^{(2)}$

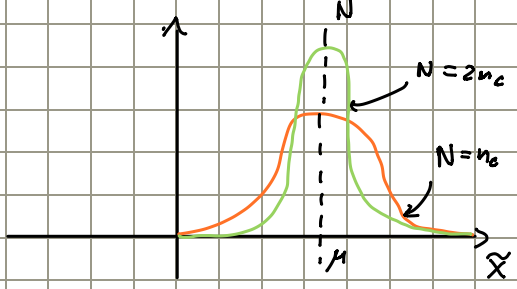


→ explain

$S_N \triangleq \frac{1}{N} \sum_{i=1}^{N} x_i \rightarrow \mu_{S_N} \triangleq E[S_N] = \frac{1}{N} E\left[\sum_{i=1}^{N} x_i\right] = \frac{1}{N} \sum_{i=1}^{N} \underbrace{E[X_i]}_{\mu} = \mu$

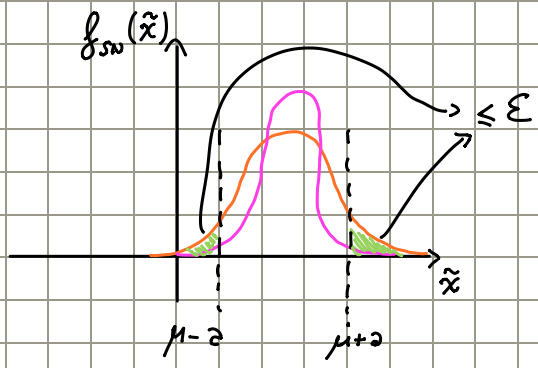
$\sigma_{S_N}^2 = \frac{\sigma_x^2}{N}$





Question from last lecture: how many samples...?

$N = ?$ s.t. $P[|S_N - \mu| > a] \leq \varepsilon$



If we know that $E[(X-\mu)^2] = \sigma_x^2 \leq S^2$

$$S_N = \frac{1}{N}\sum_{i=1}^{N} X_i \; ; \; E[S_N] = \mu$$

$$P[|S_N - \mu| > a] \leq \frac{\sigma_{S_N}^2}{a^2} = \frac{\sigma_x^2}{a^2} \leq \frac{S^2}{Na^2} \overset{!}{\leq} \varepsilon \quad \rightarrow \quad N \geq \frac{S^2}{a^2\varepsilon}$$

↑ Chebyshev bound

↑ $\sigma_x^2 \leq S^2$ by assumption

Larger variance → less predictable. We want a lot of samples

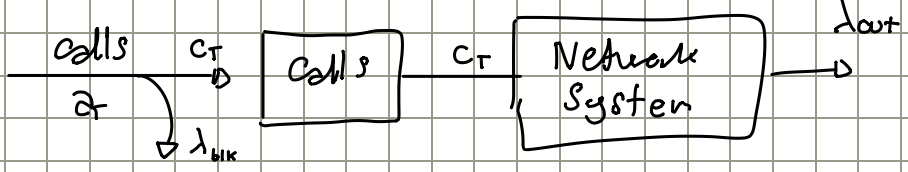
$$N \geq \frac{S^2}{a^2\varepsilon} \Rightarrow P[|S_N - \mu| > a] \leq \varepsilon' < \varepsilon$$

Example

New information flows arrive as a P.P.P. of rate $\lambda \left[\frac{\text{call}}{s}\right]$

$\forall$ call lasts for an exp. distr. time $\delta \sim \mathcal{E}(\mu)$

$\forall$ call corresponds to a CBR flow of r [bits/s]



$$P_{blk} \triangleq \frac{\lambda_{blk}}{\lambda}$$

Simulation lasts for a period T of simulated time

of arrivals: $A_T \sim P(\lambda T)$

↳ $a_T' = 131$ arrivals ← example

} if stochastic, different values.

$a_T^{(i)} = 98$)

What value do we need?

- Maybe the calls that we are storing: C_T

How do we compute the block prob.?

$$\tilde{P}_{blk} = 1 - \frac{C_T}{a_T} = \frac{b_T}{a_T} \qquad b_T = \#\text{ blocked calls},\ a_T - C_T$$

After N exp: $(a_T^{(1)}, a_T^{(2)}, \dots, a_T^{(N)})$

$(b_T^{(1)}, \dots, b_T^{(N)})$

$(\hat{P}_{blk}^1, \dots, \tilde{P}_{blk}^1)$

Q: $N = ?$ s.t.

$\tilde{P}_{blk} =$

1. $\tilde{P}_{blk} = \frac{1}{N} \sum_{x=1}^{N} \tilde{P}_{blk}^{(k)}$
2. $\tilde{P}_{blk} = \cancel{\max_{1 \le k \le N}} \tilde{P}_{blk}^{(k)}$ → explain
3. $\tilde{P}_{blk} = STD\left(\tilde{P}_{blk}^{(k)},\ k \in \{1, \dots, N\}\right)$
4. $\tilde{P}_{blk} = \frac{\sum_{k=1}^{N} b_T^{(k)}}{\sum a_T^{(k)}}$

1. Maybe won't represent well the outliers (explain better)
2. explain why
3. STD → getting rid of the mean
4. ?

1. $P_{blk} = \frac{1}{N} \cdot \left(\frac{b_T^{(1)}}{a_T^{(1)}} + \dots + \frac{b_T^{(N)}}{a_T^{(N)}} \right)$

4. $P_{blk} = \left(\frac{b_T^{(1)} + \dots b_T^{(N)}}{a_T^{(1)} + a_T^{(2)} + \dots + a_T^{(N)}} \right) = \frac{a_T^{(2)} + \alpha\left(a_T^{(2)} + a_T^{(3)} + \dots + a_T^{(N)}\right)}{a_T^{(1)} + a_T^{(2)} + \dots + a_T^{(N)}}$

1. $\tilde{P}_{blk} = \frac{1}{N} + \alpha - \frac{\alpha}{N} \xrightarrow{N \to \infty} \alpha$

2. $\tilde{P}_{blk} = \frac{a_T^{(1)}}{\sum_k a_T^{(k)}} + \frac{\alpha}{\frac{a_T^{(1)}}{\sum_{k=2}^{N} a_T^{(k)}} + 1} \xrightarrow{N \to \infty} \alpha$

$a_T^{(i)} = a_T \quad \forall i$

$= \frac{1}{N} + \frac{\alpha}{1 + \frac{1}{N-1}} = \frac{1}{N} + \frac{N-1}{N}\alpha = \frac{1}{N} + \alpha - \frac{\alpha}{N}$

Consider the first approach

$N = ? : P[\,|\tilde{P}_{blk} - E[P_{blk}]| \geq 10^{-2}\,] \leq 1\%$

$P[\,|\tilde{P}_{blk} - E[P_{blk}]| \leq 10^{-2}\,] \leq 10^{-2}\,] \geq 99\%$

The more data, the better.

Outlier: something that doesn't comply to the model

How do we find N? (the bounds)

Chebishev? No variance..

Hoeffding's Inequality

If we have $\{X_i\}$ s.t. $a_i \leq X_i \leq b_i \quad \mu_i \triangleq E[X_i] < \infty$

we consider $S_N \triangleq \frac{1}{N}\sum_{i=1}^{N} X_i \rightarrow \mu_S = E[S_N] = \frac{1}{N}\sum_{i=1}^{N} \mu_i$

$P[S_N - \mu_S \geq a] \; e^{-\frac{2a^2N^2}{\sum_{i=1}^{N}\Delta_i^2}}$ where $\Delta_i = b_i - a_i$

When $\Delta_i = \Delta \quad \forall i \quad \left(\begin{matrix} b_i = b \\ a_i = a \end{matrix}\right)$

$$P[S_N - \mu_N \geq a] \leq e^{-\frac{2a^2N}{\Delta^2}}$$

$$P[|S_N - \mu_N| \geq a] \leq 2e^{-\frac{2a^2N}{\Delta^2}}$$

Remind

$$S_N = \frac{1}{N}\sum_{k=1}^{N} X_k \qquad \text{vs} \qquad E[S_N] = \frac{1}{N}\sum_{k=1}^{N} \mu_k$$

$$\tilde{x}^{(1)} = \frac{1}{N}\sum_{k=1}^{N} x_k^{(1)}$$

$$\tilde{x}^{(2)} = \frac{1}{N}\sum_{k=1}^{N} x_k^{(2)}$$

$$\vdots$$

Why we use both? [Reply]

Proof [Hdf]

Hoeffding's Lemma

$\forall x$ r.v. with $E[x] < \infty$

$\forall \theta \in \mathbb{R}^+ \qquad E[e^{\theta(x - E[x])}] \leq e^{\frac{\theta^2\Delta^2}{8}}$ (Trust)

Hoeffding's ineq.

$x_i \in [a_i, b_i]$; $\mu_i = E[x_i]$

$S_N \triangleq \frac{1}{N}\sum_{k=1}^{N} x_i$; $\mu_S \triangleq E[S_N] = \frac{1}{N}\sum_{i=1}^{N} \mu_i$

$$P[|S_N - \mu_S| > a] \leq 2e^{-\frac{2a^2N^2}{\sum_{i=1}^{N}\Delta_i^2}}$$

Proof of Lemma

$$P[S_N - \mu \geq a] = P[\underbrace{e^{\theta(S_N - \mu_S)}}_{>0} \geq e^{\theta a}] \leq \frac{E[e^{\theta(S_N-\mu)}]}{e^{a\theta}}$$

$\theta \geq 0$

$f(\theta_x)$ is monot. increasing

$$= \frac{E[e^{\theta \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_i)}]}{e^{a\theta}} = \frac{E[\prod_{i=1}^{N} e^{\theta \frac{1}{N}(x_i-\mu_i)}]}{e^{a\theta}} = \prod_{i=1}^{N} E[e^{\frac{\theta}{N}(x_i-\mu_i)}] \cdot e^{-a\theta} \leq$$

$\{x_i\}$ are indep

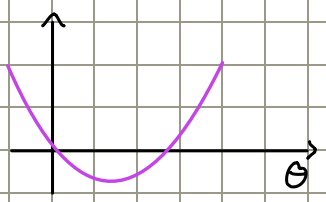
$$\leq e^{\frac{(\frac{\theta}{N})^2 \cdot \Delta_i^2}{8}} \cdot e^{-a\theta}$$

↑ Hoeff. lemma

$$= e^{\sum_{i=1}^{N} \frac{\theta^2}{N^2} \cdot \frac{\Delta_i}{8} - a\theta}$$

$$P[S_N - \mu_S > a] \leq e^{\overbrace{-a\theta + \sum_{i=1}^{N} \frac{\theta^2}{N^2} \cdot \frac{\Delta_i^2}{8}}^{\triangleq f(\theta)}}$$

$\theta > 0$



$$\frac{df(\theta)}{d(\theta)} = 0 \Rightarrow \frac{2\theta}{N^2} \cdot \sum_{i=1}^{N} \Delta_i^2 - a = 0 \Rightarrow \theta^* = \frac{aN^2}{2\sum_{i=1}^{N} \frac{\Delta_i^2}{8}} = \frac{4aN^2}{\sum_i \Delta_i^2}$$

$$= P[S_N - \mu_S > a] \leq e^{-a\theta^* + \sum_{i=1}^{N} \frac{\Delta_i^2}{8} \cdot \frac{\theta^{*2}}{N^2}}$$

$$= \dots = e^{-\frac{2N^2a^2}{\sum_{i=1}^{N} \Delta_i^2}}$$

$$P[\mu_S - S_N > a] \leq e^{-\frac{2N^2a^2}{\sum_{i=1}^{N} \Delta^2}} \leq \varepsilon$$